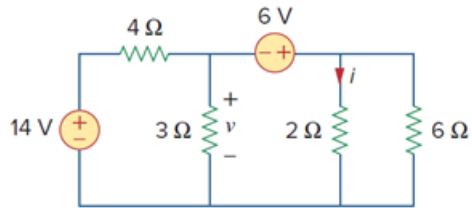


Problem

Find v and i in the circuit of Fig. 3.11.

Figure 3.11:



Step-by-step solution

Step 1 of 6

Draw the circuit diagram.

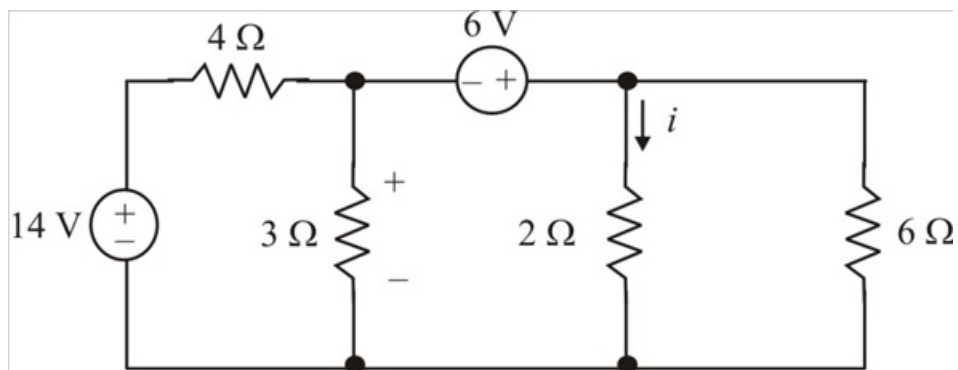


Figure 1

Step 2 of 6

Draw the circuit diagram designating with currents and voltages.

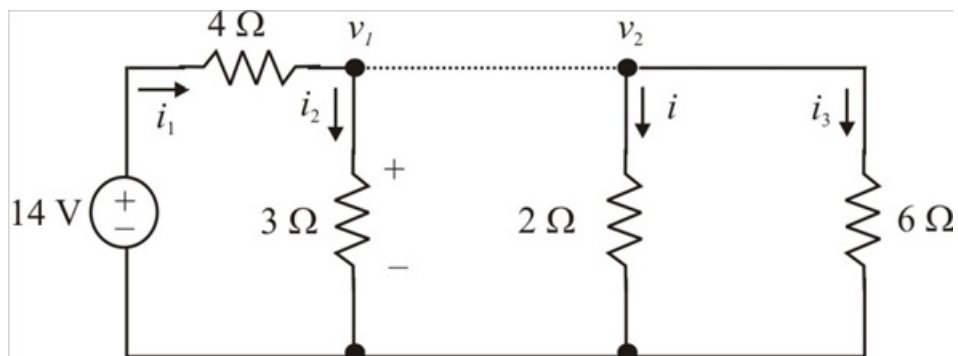


Figure 2

Step 3 of 6

The super node contains the 6 V source, node 1 and node 2. Apply Kirchhoff's current law to the super node as shown in Figure 2.

$$i_1 = i_2 + i + i_3$$

$$\frac{14 - v_1}{4} = \frac{v_1}{3} + \frac{v_2}{2} + \frac{v_2}{6}$$

Multiplying each term in the last equation by 12, we obtain

$$\frac{14 - v_1}{4} = \frac{v_1}{3} + \frac{v_2}{2} + \frac{v_2}{6}$$

$$42 - 3v_1 = 4v_1 + 6v_2 + 2v_2$$

$$7v_1 + 8v_2 = 42 \dots\dots (1)$$

Step 4 of 6

Draw the modified circuit diagram.

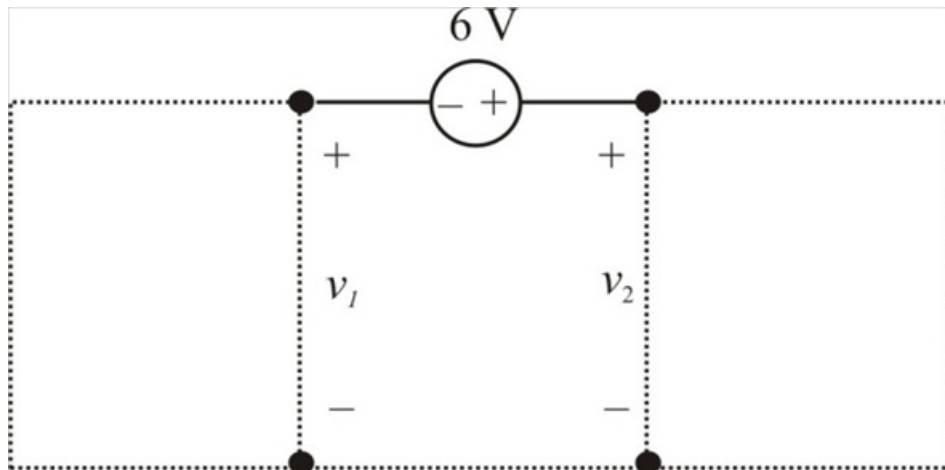


Figure 3

Step 5 of 6

To get the relation between v_1 and v_2 , we apply Kirchhoff's voltage law to the circuit in Figure 3.

$$-v_1 - 6 + v_2 = 0$$

$$-v_1 + v_2 = 6$$

Multiplying each term by 7, we obtain

$$-7v_1 + 7v_2 = 42 \dots\dots (2)$$

Add equation (1) and equation (2).

$$7v_1 + 8v_2 = 42$$

$$\frac{-7v_1 + 7v_2 = 42}{15v_2 = 84}$$

$$15v_2 = 84$$

$$v_2 = 5.6 \text{ V}$$

Applying Ohm's law,

$$i = \frac{v_2}{R}$$

$$= \frac{5.6}{2}$$

$$= 2.8 \text{ A}$$

Therefore the current i is 2.8 A.

Step 6 of 6

Substitute the value of v_2 in equation (1).

$$7v_1 + 8v_2 = 42$$

$$7v_1 + 8(5.6) = 42$$

$$7v_1 = -2.8$$

$$v_1 = -0.4 \text{ V}$$

But,

$$v_1 = v$$

$$= -0.4 \text{ V}$$

$$= -400 \text{ mV}$$

Therefore the voltage v is -400 mV.